FINALTERM EXAMINATION
Spring 2010
CS610- Computer Network

Ref No: 1578657
Time: 90 min
Marks: 60

Question No: 1 (Marks: 1) - Please choose one

A ---------- relies on the hardware manufacturer to assign a unique physical address to each network interface.

► **Static addressing scheme**
► Configurable addressing scheme
► Dynamic addressing scheme
► None of the given

Question No: 2 (Marks: 1) - Please choose one

An interface for thin Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

► **RJ-45, 10 Base T**
► RJ-45, 10 Base 5
► BNC, 10 Base 2
► BNC, 10 Base T

Question No: 3 (Marks: 1) - Please choose one

A system with redundant bridges might have a problem with_____ in the system.

► **Loop**
► Filters
► Spanning Trees
► All given choices

Question No: 4 (Marks: 1) - Please choose one

A Bridge can _____

► Filter a frame
► Forward a frame
► Extend a LAN
► **Do all the above**

Question No: 5 (Marks: 1) - Please choose one

_____ is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

► Constant Bit Rate (CBR) service
► Variable Bit Rate (VBR) service
► **Available Bit Rate (ABR) service**
► None of the given

Question No: 6 (Marks: 1) - Please choose one

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

► 21-bit
► 22-bit
► 23-bit
► **24-bit**

Question No: 7 (Marks: 1) - Please choose one

_____ of TCP/IP layering model, corresponds to basic network hardware.

► **Physical Layer**

► Network Interface Layer

► Internet Layer

► Transport Layer

Question No: 8 (Marks: 1) - Please choose one

_____ places the boundary between the second and third octets

► **Class A**

► Class B

► Class C

► Class D

Question No: 9 (Marks: 1) - Please choose one

UDP and TCP are both_____ layer protocols

► Physical

► Data link

► Network

► **Transport**

Question No: 10 (Marks: 1) - Please choose one

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by_____

► IP

► None of the given

► **TCP**

► UDP

Question No: 11 (Marks: 1) - Please choose one

_____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer

► Network Interface Layer

► Internet Layer

► **Transport Layer**

Question No: 12 (Marks: 1) - Please choose one

_____identifies which application program on receiving computer should receive the data

► Logical address

Source port

Logical address

None of the give

► Source port

► **Destination Port**

► None of the given

Question No: 13 (Marks: 1) - Please choose one

_____ identifies the application program that sent the data.

► Destination Port

► Source port

► Logical address

► None of the given

Question No: 14 (Marks: 1) - Please choose one

Which of the following are interior routing protocols?

► RIP

► OSPF

► BGP

► RIP and OSPF

Question No: 15 (Marks: 1) - Please choose one

The Border Gateway Protocol (BGP) uses_____ for all communication

► UDP

► TCP

► Both UDP and TCP

► None of the given

Question No: 16 (Marks: 1) - Please choose one

_____measures distance in network hops, where each network between the source and destination counts as single hop.

► BGP

► OSPF

► RIP

► None of the given

Question No: 17 (Marks: 1) - Please choose one

OSPF is based on--------

► Distance vector routing

► Link state routing

► Path vector routing

► Distance vector routing and Link state routing

Question No: 18 (Marks: 1) - Please choose one

_____ performs local multicast and uses IP-in-IP encapsulation to send multicast datagrams from one site on the Internet to another.

► Distance Vector Multicast Routing Protocol (DVMRP)

► Core Based Trees (CBT)

► Protocol Independent Multicast_ *Sparse Mode (PIM-SM)*

► Protocol Independent Multicast _ *Dense Mode (PIM-DM)*

Question No: 19 (Marks: 1) - Please choose one

The length of time required to send a variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.

► True

► False

Question No: 20 (Marks: 1) - Please choose one

NEXT HEADER field in the base header defines type of header and it appears at end of fixed-size base header.

► True

► False

Question No: 21 (Marks: 1) - Please choose one

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

► **True**

► False

Question No: 22 (Marks: 1) - Please choose one

Address mask defines how many bits of address are in suffix?

► True

► **False**

Question No: 23 (Marks: 1) - Please choose one

A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?

► **True**

► False

Question No: 24 (Marks: 1) - Please choose one

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False

► **True**

Question No: 25 (Marks: 1) - Please choose one

The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address_____.

► **itself**

► prefix

► suffix

► mask

Question No: 26 (Marks: 1) - Please choose one

In which method of Address Resolution Protocol the protocol address independent of hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T, C

► D

► C

► **T, D**

Question No: 27 (Marks: 1) - Please choose one

In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T
► D
► **C**
► T, C

Question No: 28 (Marks: 1) - Please choose one

Reconstruction of original datagram is called reassembly.

► **True**
► False

Question No: 29 (Marks: 1) - Please choose one

A computer needs a complete stack of protocols to run either a client or a server.

► **True**
► False

Question No: 30 (Marks: 1) - Please choose one

TCP uses _____ mechanism to control the flow of data.

► door
► **window**
► acknowledgment
► retransmission

Question No: 31 (Marks: 2)

To solve the problem the routers and networks in the Internet are divided into groups. How large do you think are these groups?

Question No: 32 (Marks: 2)

Traceroute continues to increment the Time To Live until the value is large enough for the datagram to reach its final destination. What happens when the TTL is sufficiently large for the datagram to reach its destination?

Question No: 33 (Marks: 2)

How can TCP monitor internet delays?

Question No: 34 (Marks: 2)

Which wireless standard is used in WIFI technology?

Question No: 35 (Marks: 3)

How does IP software reassemble fragments that arrive out of order?

Question No: 36 (Marks: 3)
What kind of information is in a routing table?

Question No: 37 (Marks: 3)
Define frames without type field?

Question No: 38 (Marks: 3)
What is the meaning of *Facilities For Transit Routing* as a characteristic of the Border Gateway Protocol?

Question No: 39 (Marks: 5)
What are the charachteristics of the Border Gateway Protocol?

Question No: 40 (Marks: 5)
Describe how NAT can be usefull for residence with cable modem or DSL?

FINALTERM EXAMINATION
Spring 2010
CS610- Data Communication

Question No: 1 (Marks: 1) - Please choose one
_____representation of links that connect nodes is called as physical topology.
► geometrical
► logical
► physical

Question No: 2 (Marks: 1) - Please choose one
The internet model consists of _____ layers.
► three
► two
► five
► seven

Question No: 3 (Marks: 1) - Please choose one
Encryption and encoding are the same terms.
True
► False
Question No: 4 (Marks: 1) - Please choose one
he amplitude of a digital signal depends upon the_____ to represent a bit.
► phase
► voltage
► wavelength
Question No: 5 (Marks: 1) - Please choose one
The inversion of the level at 1 bit is called as _____
►NRZ-L
► NRZ-I
► RZ
Question No: 6 (Marks: 1) - Please choose one
Modulation of an analog signal can be accomplished through changing the _____ of the carrier signal.
► amplitude
► frequency
► phase
► all of the given
Question No: 7 (Marks: 1) - Please choose one
If FCC regulations are followed, the carrier frequencies of adjacent AM radio stations are _____ apart.
► 5 KHz
► 10 KHz
► 200 KHz
► 530 KHz
Question No: 8 (Marks: 1) - Please choose one
Category 5 UTP cable is used for data transmission of upto_____.
► 100 Mbps
► 200 Mbps
► 250 Mbps
► 400 Mbps
Question No: 9 (Marks: 1) - Please choose one
The RG number gives us information about _____.
► Twisted pairs
► Coaxial cables
► Optical fibers
► all of the given
Question No: 10 (Marks: 1) - Please choose one
The inner core of an optical fiber is _____ in composition.
► Glass plastic
► Copper
► Bimetallic
► Liquid

Question No: 11 (Marks: 1) - Please choose one
All of popular Fiber optic connectors are _____ shaped.
► Conical
► Barrel
► Circular
► Rectangular
Question No: 12 (Marks: 1) - Please choose one
The VLF and LF bands use _____ propagation for communications.
► Ground
► Sky
► Line of sight
► Space
Question No: 13 (Marks: 1) - Please choose one
Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across _____ data link
► Single
► Multi
► Single and Multi
► none of the given
Question No: 14 (Marks: 1) - Please choose one
A portion of the path that carries TX b/w a given pair of devices is known as _____.
► Node
► Bridge
► Channel
► Gateway
Question No: 15 (Marks: 1) - Please choose one
Which error detection method involves polynomials?
► Checksum
► Two-dimensional parity check
► CRC
► Simple parity check
Question No: 16 (Marks: 1) - Please choose one
If the ASCII character G is sent and the character D is received, what type of error is this?
► Single-bit
► Multiple-bit
► Burst
► Recoverable
Question No: 17 (Marks: 1) - Please choose one
Which error detection method involves the use of parity bits?
► Simple parity check & two dimensional parity check
► CRC
► Two-dimensional parity check
► Simple parity check
Question No: 18 (Marks: 1) - Please choose one

Which error detection method can detect a single-bit error?
► Simple parity check
► Two-dimensional parity check
► CRC
► All of the given

Question No: 19 (Marks: 1) - Please choose one

The Hamming code is a method of _____
► Error detection
► Error correction
► Error ecapsulation
► Error detection & Error encapsulation

Quetion No: 20 (Marks: 1) - Please choose one

Sliding window requires that data frames be transmitted _____
► Sequentially
► Frequently
► Synchronously
► Asynchronously

Question No: 21 (Marks: 1) - Please choose one

In selective-reject ARQ, only the specific damaged or lost frame is_____.
► Retransmitted
► Forwarded
► Selected
► Rejected

Question No: 22 (Marks: 1) - Please choose one

Which of the following sublyer, resolves the contention for the shared media
► MAC
► LLC
►Physical

Question No: 23 (Marks: 1) - Please choose one

The PDU has no flag fields, no CRC, and no station address
► TRUE
► FALSE

Question No: 24 (Marks: 1) - Please choose one

IEEE divides the base band category into _____ standards.
► 5
► 4
► 3
► 6

Question No: 25 (Marks: 1) - Please choose one

Like 10 Base 5, 10 Base 2 is a _____ topology LAN
► Ring
► Mesh
► Star
► Bus

Question No: 26 (Marks: 1) - Please choose one

Check sum method is used for _____ layers.

► Physical
► Application
► Transport
► Datalink

Question No: 27 (Marks: 1) - Please choose one

Repeater works on _____ layer.

► Data Link
► Physical
► Network
► Application

Question No: 28 (Marks: 1) - Please choose one

Trunks are transmission media such as _____ that handle the telephone to the nearest end office.

► Satellite links
► Twisted-pair & Fiber-optic
► Twisted-pair
► Fiber-optic

Question No: 29 (Marks: 1) - Please choose one

Which of the following _____ uses a series of filters to decompose multiplexed signal into its constituent signals.

MUX
► DEMUX
► Switch
► Bridge

Question No: 30 (Marks: 1) - Please choose one

In Fast Ethernet, data rate can be increased by _____ collisions.

► Increasing
► Decreasing
► Keeping Constant
► None of the given

Question No: 31 (Marks: 2)

What are the advantages of a multipoint connection over a point-to-point connection?

Answer:

Point-to-point connection is limited to two devices, where else more than two devices share a single link in multipoint connection. Multipoint connection can be used for fail-over and reliability.

Question No: 32 (Marks: 2)

What's the name of the telephone service in which there is no need of dialing.

Answer:

\DSS (digital data service) is the telephone service in which there is no need of dialing.

Question No: 33 (Marks: 2)

Which type of frames are present in BSC frames?

Answer:

There are two types of frames that are present in BSC.

1. Control Frames and

2. Data Frames

Question No: 34 (Marks: 2)

What methods of line discipline are used for peer to peer and primary secondary communication?

Answer:

Line discipline is done in two ways:

1. ENQ/ACK (Enquiry Acknowledgement)

This is used for peer to peer communication.

2. Poll/ Select

This method is used for primary secondary communication.

Question No: 35 (Marks: 3)

How does the checksum checker know that the received data unit is undamaged? [3]

Answer:

Checksum Checker or generator:

The sender subdivides data units into equal segments of 'n' bits(16 bits)

1. These segments are added together using one's complement.

2. The total (sum) is then complemented and appended to the end of the original data unit as redundancy bits called CHECKSUM.

3. The extended data unit is transmitted across the network.

4. The receiver subdivides data unit and adds all segments together and complement the result.

5. If the intended data unit is intact, total value found by adding the data segments and the checksum field should be zero.

6. If the result is not zero, the packet contains an error & the receiver rejects it.

Question No: 36 (Marks: 3)

Which one has more overhead, a repeater or a bridge? Explain your answer. [3]

Answer:

A bridge has more overhead than a repeater. A bridge processes the packet at two layers ; a repeater processes a frame at only one layer. A bridge needs to search a table and find the forwarding port as well as to regenerate the signal; a repeater only regenerates the signal. In other words, a bridge is also a repeater (and more); a repeater is not a bridge.

Question No: 37 (Marks: 3)

Write down disadvantages of Ring Topology.

Answer:

Disadvantages of Ring Topology

Unidirectional Traffic

A break in a ring that is a disabled station can disable the entire network

Can be solved by using:

Dual Ring or

A switch capable of closing off the Break

Question No: 38 (Marks: 3)

How parity bits are counted in VRC error detection method technique in case of odd parity generator?

Answer:

For example:

We want to TX the binary data unit 1100001
Adding together the number of 1's gives us 3, an odd number Before TX, we pass the data unit through a parity generator, which counts the 1's and appends the parity bit (1) to the end The total number of 1's is now 4, an even number The system now transfers the entire expanded across the network link When it reaches its destination, the RX puts all 8 bits through an even parity checking function
If the RX sees 11100001, it counts four ones, an even number and the data unit passes When the parity checker counts the 1's, it gets 5 an odd number The receiver knows that an error has occurred somewhere and therefore rejects the whole unit Some systems may also use ODD parity checking The principal is the same as even parity

Question No: 39 (Marks: 5)

How are lost acknowledgment and a lost frame handled at the sender site? [5]

Question No: 40 (Marks: 5)

Explain Protocol Data Unit (PDU)?

Answer: Protocol data unit (PDU) is an OSI term that refers generically to a group of information added or removed by a particular layer of the OSI model. In specific terms, an LxPDU implies the data and headers defined by layer x. Each layer uses the PDU to communicate and exchange information. The PDU information is only read by the peer layer on the receiving device and then stripped off, and data is handed over to the next upper layer.

FINALTERM EXAMINATION

Fall 2008

CS610- Computer Network

Ref No: 325806
Time: 120 min
Marks: 70

Question No: 1 (Marks: 1) - Please choose one

Computer networks are often called ---------- because they use packet technology.

- ► Ethernet
- ► Switch networks
- ► Packet networks
- ► **None of the given**

Question No: 2 (Marks: 1) - Please choose one

A network uses a ------arranges for computers to be connected in a closed loop.

- ► Star Topology
- ► **Ring Topology**
- ► Bus Topology
- ► None of the given

In a ring topology each computer is connected to other thorough a ring

Question No: 3 (Marks: 1) - Please choose one

An ----------method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

- ► **Explicit frame type**

► Ideal frame type
► Implicit frame type
► None of the given

EXPLICIT FRAME TYPE:
In this type the identifying value is included with frame describes types of included data.

Question No: 4 (Marks: 1) - Please choose one

An interface for thin Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

► RJ-45, 10 Base T (reference not available)
► RJ-45, 10 Base 5
► BNC, 10 Base 2
► BNC, 10 Base T

Question No: 5 (Marks: 1) - Please choose one

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

► Layer 2 source address
► Source node's physical address
► Layer 2 destination address (not confirm)
► Layer 3 destination address

Question No: 6 (Marks: 1) - Please choose one

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

► Hierarchal address
► Default route
► Shortest path
► None of the given

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called Default route. The mechanism allows a single entry in a forwarding table to replace a long list of entries that have the same next hope value

Question No: 7 (Marks: 1) - Please choose one

_____ of TCP/IP layering model, corresponds to basic network hardware.

► Physical Layer
► Network Interface Layer
► Internet Layer
► Transport Layer

LAYER 1:
Corresponds to basic network hardware layer in OSI

Question No: 8 (Marks: 1) - Please choose one

_____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer
► Network Interface Layer
► Internet Layer
► Transport Layer

The Internet layer in the TCP/IP reference model is responsible for transferring data between the source and destination computers

Question No: 9 (Marks: 1) - Please choose one

_____is called an end-to-end protocol because it provide a connection directly from an application on one computer to an application on a remote computer.

► IP

► UDP

► TCP

► None of the given

UDP has the following characteristics.

It is an end-to-end protocol. It provides application-to-application communication.

Question No: 10 (Marks: 1) - Please choose one

_____ uses distance vector approach to define routing

► BGP

► OSPF

► RIP

► None of the given

DISTANCE-VECTOR ROUTING:

• It is very simple to implement.

• Packet switch updates its own routing table first.

• It is used in RIP.

Question No: 11 (Marks: 1) - Please choose one

_____ is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

► Flood-and -Prune

► Configuration-and -Tunneling

► Core-Based Discovery

► None of the give006E

FLOOD-AND-PRUNE:

Flood-and-prune is ideal in a situation where the group is small and all members Are attached to contiguous Local Area Networks.

Question No: 12 (Marks: 1) - Please choose one

Router that decrements TTL to __ sends ICMP time exceeded message, with router's address as source address

► 3

► 2

► 1

► 0

Router that decrements TTL to 0 sends ICMP time exceeded message, with router's address as source address

Question No: 13 (Marks: 1) - Please choose one

Protocol addresses are abstractions provided by _____.

► hardware

► software

► operating system

► internet

PROTOCOL ADDRESSES AND PACKET DELIVERY:

Protocol addresses are abstractions provided by software

Question No: 14 (Marks: 1) - Please choose one

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

► **True**

► False

CASHING ARP RESPONSES:

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient

Question No: 15 (Marks: 1) - Please choose one

ARP is almost always used to bind a ___-bit IP address to a ___-bit Ethernet address.

► **32, 48**

► 24, 32

► 32, 64

► 32, 128

ARP is almost always used to bind a 32-bit IP address to a 48-bit Ethernet address

Question No: 16 (Marks: 1) - Please choose one

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False

► **True**

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

Question No: 17 (Marks: 1) - Please choose one

In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T

► D

► **C**

► T, C

Features	**Types of Resolution**
Use full with any hardware	T
Address change affects all hosts	T
Protocol address is determined by hardware address	C

Question No: 18 (Marks: 1) - Please choose one

The general form of an IP datagram is with a header followed by data. The header contains information that controls where and how the datagram is to be sent.

► **True**

► False

- Header contains all information needed to deliver datagram to the destination

computer. It contains:
• Destination address
• Source address
• Identifier
• Other delivery information

Question No: 19 (Marks: 1) - Please choose one

To save traffic, an EGP does not summerize routing information from the autonomous system before passing it to another autonomous system.

► True
► False

"It provides routing among autonomous systems (EGP).

Question No: 20 (Marks: 1) - Please choose one

Which of the following is a correct representation of the IPv6?

► 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255
► 105.220.136.100.255.255.255.256.0.0.18.128.140.10.255.255
► 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255.256
► 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255

128-bit addresses unwidely in dotted decimal; requires 16 numbers:
105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255

Question No: 21 (Marks: 2)

Is there a comparison between TCP/IP reference model and OSI reference model.

Both the TCP/IP and OSI are international standards for data communication. Both use the concept of protocol layering. These are theconceptual tools that is often used to show how various protocols and technologies fit together to implement networks. The functions that the model represents are much the same. This means that the TCP/IP and the OSI models are really quite similar in nature even if they don't carve up the network functionality pie in precisely the same way. There is a fairly natural correspondence between the TCP/IP and OSI layers; it just isn't always a "one-to-one" relationship. Since the OSI model is used so widely, it is common to explain the TCP/IP architecture both in terms of the TCP/IP layers and the corresponding OSI layers

Question No: 22 (Marks: 2)

How can a datagram be transmitted across a physical network that does not understand the datagram format?

Encapsulate!

- The network interface layer encapsulates an entire datagram in the data area of a hardware frame.
- The network hardware ignores the datagram format.
- The hardware treats a frame containing a datagram like any other frame.
- The sender places a frame type in the frame header indicating that the frame contains a datagram.
- The receiver reads the frame type and knows the frame contains a datagram.

Question No: 23 (Marks: 2)

What is the basic concept of Twice NAT (Network Address Translation)?

Twice NAT is another variant of NAT. it is used with site that runs server. In this process NAT box is connected to Domain Name.

Question No: 24 (Marks: 3)

What format is used for an internet packet?

Question No: 25 (Marks: 3)

"To achieve a hierarchy, OSPF allows an autonomous system to be partitioned for routing purposes". Does this feature make OSPF more complex or powerful?

Question No: 26 (Marks: 3)

Why does IPv6 use separate Extension Headers?

IPv6 use separate Extension Headers. Fragmentation information is kept in separate extension header. Each fragment has base header and (inserted) fragmentation header. Entire datagram including originalheader may be fragmented.

Question No: 27 (Marks: 5)

Consider the **IP** addresses: **178.200.127.5** and the corresponding **subnet masks 255.255.255.0,** then find out the following:

a. The number of bits used for subnetting (Answer **8 bits)**
b. Total number of host in the subnet (Answer **254)**
c. The network address of the subnet. (Answer **178.200.127.0)**
d. The subnet address of the IP address.

Here is the online calculator for subnetting

http://www.subnetmask.info/
http://www.subnet-calculator.com/subnet.php?net_class=B

NOTE. This is just for concept not included in the answer

The subnet mask determines what portion of the TCP/IP address represents your network and what portion can be used for your hosts. The network number represents the street I live on, and the host portion is used for the numbers on all the houses on my street.
A subnet mask of **255.255.255.0** means that the first three octets of the address will be used for the network, and thus our network number is **192.168.1**. This means we can have **254** computers on this network, because the fourth octet is not being used by the network portion of the address. We know this because of the **0** in the subnet mask (255.255.255.**0**).

Question No: 28 (Marks: 5)

How does IP software reassemble fragments that arrive out of order?

A unique ID number of each outgoing datagram

- When a router fragments the datagram, the router copies the ID number into each fragment.
- The fragment OFFSET field tells a receiver how to order fragmentswithin a given datagram
- The process of creating a copy of the original datagram from fragments is called reassembly.

Question No: 29 (Marks: 5)

Write down the comparison of Distance- vector and Link – state algorithm?

COMPARISON:

DISTANCE-VECTOR ROUTING:

• It is very simple to implement.
• Packet switch updates its own routing table first.
• It is used in RIP.

LINK-STATE ALGORITHM:

• It is much more complex.

• Switches perform independent computations.
• It is used in OSPF.

Question No: 30 (Marks: 10)

Descibe in detail what is the purpose of the following table? What sort of information can be extracted?

First Four Bits Of address	Table index in decimal	Class of Address
0000	0	A
0001	1	A
0010	2	A
0011	3	A
0100	4	A
0101	5	A
0110	6	A
0111	7	A
1000	8	B
1001	9	B
1010	10	B
1011	11	B
1100	12	C
1101	13	C
1110	14	D
1111	15	E

The table shows in the figure, how the class of address can be computed.

Question No: 31 (Marks: 10)

List down and describe atleast five charachteristics of Routing Information Protocol.

ROUTING INFORMATION PROTOCOL (RIP):

It has the following characteristics:

"It is used for routing within an autonomous system (IGP).

"Hop Count Metric: RIP measures distance in network hops, where each network between the source and destination counts as a single hop.

"It uses UDP for all message transmissions.

"RIP is used over LAN. Version 1 of RIP uses hardware broadcast and version 2 allows delivery via multicast.

"It can be used to advertise default route propagation. An organization can use RIP to install a default route in each router.

"It uses distance vector algorithm.

"RIP allows hosts to listen passively and update its routing table

FINALTERM EXAMINATION

Fall 2008

CS610- Computer Network

Ref No: 325624

Time: 120 min

Marks: 70

Question No: 1 (Marks: 1) - Please choose one

The number of connections needed for N computer in direct point to point communication is equal to:

► **$(N^2–N)/2$**

► N(N- 1)

► N^2

► None of the given

For 'n' computers connections in direct point to point communication = (n^2 –n)/2.

Question No: 2 (Marks: 1) - Please choose one

When an application---------- data, it makes a copy of the data available to all other computers on the network.

► Broadcasting

► Multicasting

► Unicasting

► **None of the given (not sure)**

Question No: 3 (Marks: 1) - Please choose one

Ethernet uses a ------ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

► 64

► **48**

► 32

► 8

Ethernet uses a 48-bit static addressing scheme

Question No: 4 (Marks: 1) - Please choose one

A system with redundant bridges might have a problem with_____ in the system.

► **Loop**

► Filters

► Spanning Trees

► All given choices

Reference not available

Question No: 5 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

► Area

► Volume

► Length

► None of the given

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network

Question No: 6 (Marks: 1) - Please choose one

Connectionless service, Message-Oriented protocol, best effort delivery service, arbitrary interaction and operating system independent are the characteristics of _____

► TCP

► **UDP**

► IP
► None of the given

UDP has the following characteristics.
• It is an end-to-end protocol. It provides application-to-application communication.
• It provides connectionless service.
• It is a Message-Oriented protocol.
• It uses best-effort delivery service.
• It follows arbitrary interaction.
• It is operating system independent

Question No: 7 (Marks: 1) - Please choose one

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by_____

► None of the given
► **TCP**
► UDP
► IP

SERVICE PROVIDED BY TCP:
Following are the services provided by TCP:
• Connection-oriented service
• Point-to-point
• Complete reliability
• Full-duplex communication
• Stream interface
• Reliable connection startup
• Graceful connection shutdown

Question No: 8 (Marks: 1) - Please choose one

The process of using a routing table to select a next hop for a given datagram is called_____

► Encapsulation
► Reassembling
► **Routing or forwarding**
► None of the given

Reference not available

Question No: 9 (Marks: 1) - Please choose one

_____ uses distance vector approach to define routing

► BGP
► OSPF
► **RIP**
► None of the given

DISTANCE-VECTOR ROUTING:
• It is very simple to implement.
• Packet switch updates its own routing table first.
• It is used in RIP.

Question No: 10 (Marks: 1) - Please choose one

A multicast routing scheme in which the protocol software builds a delivery tree from a central point is called_____

► Distance Vector Multicast Routing Protocol (DVMRP)

► **Core Based Trees (CBT)**

► Protocol Independent Multicast_ *Sparse Mode (PIM-SM)*

► Protocol Independent Multicast _ *Dense Mode (PIM-DM)*

CORE BASED TREES (CBT):

A multicast routing scheme in which the protocol software builds a delivery tree from a central point.

Question No: 11 (Marks: 1) - Please choose one

One repeater _____, two repeaters _____ the maximum cable length limitation.

► doubles, cancel

► **doubles, triple**

► square roots, cude roots

► and, triple

One repeater doubles, two repeaters triple the maximum cable length limitation.

Question No: 12 (Marks: 1) - Please choose one

Whenever it handles a packet, IP software needs to separate the destination address into a _____ and _____.

► postfix, Infix

► non of these

► Infix, prefix

► **prefix, suffix**

Whenever it handles a packet, IP software needs to separate the destination address into a prefix and suffix

Question No: 13 (Marks: 1) - Please choose one

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

► **True**

► False

CASHING ARP RESPONSES:

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

Question No: 14 (Marks: 1) - Please choose one

ARP is almost always used to bind a ___-bit IP address to a ___-bit Ethernet address.

► **32, 48**

► 24, 32

► 32, 64

► 32, 128

ARP is almost always used to bind a 32-bit IP address to a 48-bit Ethernet address

Question No: 15 (Marks: 1) - Please choose one

End-to-end delivery service is connection oriented.

► True

► **False**

End-to-end delivery service is connection less

Question No: 16 (Marks: 1) - Please choose one
A single networking technology is best for all needs.

► True

► False

There is no single networking technology that is best for all needs
Question No: 17 (Marks: 1) - Please choose one
In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.
► False
► True
In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.
Question No: 18 (Marks: 1) - Please choose one
Which method of Address Resolution Protocol is useful with any hardware?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?
► T
► C
► D
► C, D

Features	Types of Resolution
Use full with any hardware	T
Address change affects all hosts	T
Protocol address is determined by hardware address	C

Question No: 19 (Marks: 1) - Please choose one
In which method of Address Resolution Protocol the protocol address is determined by hardware address?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?
► T
► D
► C
► T, C

Features	Types of Resolution
Use full with any hardware	T
Address change affects all hosts	T
Protocol address is determined by hardware address	C

Question No: 20 (Marks: 1) - Please choose one
We use the term _____ to refer to a measure of the path that routing software use when choosing a route.

► routing path
► **routing metric**
► routing
switching

We use the termrouting metric to refer to a measure of the path that routing software use when choosing a route

Question No: 21 (Marks: 2)

Is there a comparison between TCP/IP reference model and ISO reference model.

Question No: 22 (Marks: 2)

Does OSPF only share information within an area or does it allow communication between areas?

OSPF allows subdivision of Autonomous System into areas. The link-status information is propagated within an area. The routes are summarized before being propagated to another area. It reduces overhead (less broadcast traffic). Because it allows a manager to partition the routers and networks in an autonomous system into multiple areas, OSPFcan scale to handle a larger number of routers than other IGPs.

Question No: 23 (Marks: 2)

What are the implementations of Network Address Translation?

Question No: 24 (Marks: 3)

Describe the difference between static and dynamic routing?

STATIC ROUTING:

It is one of the forms of Internet routing. In Static routing, the table is initialized when system boots and there is no further changes.

DYNAMIC ROUTING:

In dynamic routing the table is initialized when system boots. It includes routing software which learns routes and updates table. In this way continuous changes are possible due to routing software.

Question No: 25 (Marks: 3)

What is the first address in the block if one of the addresses is 140.120.84.24/20?

The first address in the block is 140.120.80.0/20

WE can use the following short cut to find the first address

1. divide the prefix length into four groups(corresponding to the four bytes in an address) and find the number of 1s in each group
2. if the number of 1s in a group is 8 the corresponding byte in the first address is the same (no change)
3. If the number of 1s in the group is zero (no1s), the corresponding byte in the first address is 0
4. if the number of 1s in a group is between zero and eight, we keep the corresponding bits in that group

Question No: 26 (Marks: 3)

Write three new features of IPV6.

The new features of IPV6 are as follows:

• IPV6 addresses are 128 bits.
• Header format is entirely different.
• Additional information is stored in optional extension headers, followed by data.

• Flow label and quality of service allows audio and video applications to establish appropriate connections.
• New features can be added more easily. So it is extensible.

Question No: 27 (Marks: 5)

What is the difference between an interior gateway protocol and an exterior gateway protocol? Name an example of each.

INTERIOR GATEWAY PROTOCOLS (IGPs):

It is used among routers within autonomous system. The destinations lie within IGP.

EXTERIOR GATEWAY PROTOCOLS (EGPs):

It is used among autonomous systems. The destinations lie throughout Internet

Question No: 28 (Marks: 5)

As the Internet grew, the original Classful addressing scheme became a limitation, what is the designed solution.

Question No: 29 (Marks: 5)

What are IPv6 ADDRESS NOTATION?

IPv6 ADDRESS NOTATION:

128-bit addresses unwidely in dotted decimal; requires 16 numbers:
105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255
Groups of 16-bit numbers in hex separated by colons – colon hexadecimal (or colon hex).
69DC: 8864:FFFF: FFFF: 0:1280:8C0A:FFFF
Zero-compression – series of zeroes indicated by two colons
FF0C: 0:0:0:0:0:0:B1
FF0C::B1
IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address.

Question No: 30 (Marks: 10)

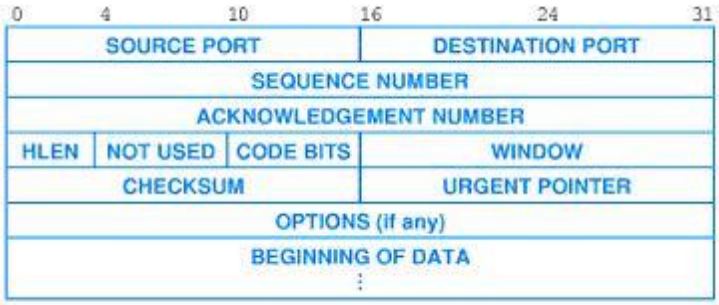
LIST SOME CHARACTERISTICS OF A CLIENT.

CHARACTERISTICS OF A CLIENT:

The characteristics of a client are explained below:
"Client is an arbitrary application program.
"It becomes client temporarily.
"It can also perform other computations.
"It is invoked directly by the user.
"It runs locally on the user's computer.
"It actively initiates contact with a server.
"It contacts one server at a time.

Question No: 31 (Marks: 10)

With help of the diagram below, explain TCP Segment Format:



TCP uses single format for all messages. TCP uses the term segment to refer to a message. Each message sent from TCP on onemachine to TCP on another machine uses this format including data andacknowledgement

Field Name	Size (bytes)	Description
SourcePort	2	ephemeral (client) port number**SourcePort:** The 16-bit port number of the process that originated the TCP segment on the source device. This will normally be an for a request sent by a client to a server
DestinationPort	2	***DestinationPort:*** The 16-bit port number of the process that is the ultimate intended recipient of the message on the destination device.
Sequence Number	4	***Sequence Number:*** For normal transmissions, the sequence number of the first byte of data in this segment. In a connection request (*SYN*) message, this carries the initial sequence number (ISN) of the source TCP.
Acknowledgment Number	4	***Acknowledgment Number:*** When the *ACK* bit is set, this segment is serving as an acknowledgment (in addition to other possible duties) and this field contains the sequence number the source is next expecting the destination to send.
Window	2	***Window:*** Indicates the number of octets of data the sender of this segment is willing to accept from the receiver at one time. This normally corresponds to the current size of the buffer allocated to accept data for this connection.
Checksum	2	16-bit checksum for data integrity protection***Checksum:*** A , computed over the entire TCP datagram plus a special “pseudo header” of fields. It is used to protect the entire TCP segment against not just errors in transmission, but also errors in delivery.
Urgent Pointer	2	***Urgent Pointer:*** Used in conjunction with the *URG* control bit for priority data transfer. This field

		contains the sequence number of the last byte of urgent data.
Options	Variable	***Options***: TCP includes a generic mechanism for including one or more sets of optional data in TCP segment. Each of the options can be either one byte in length or variable length. The first byte if the option kind subfield and it's value specifies the type of option, which in turn indicates whether the option is just a single byte or multiple bytes. Options that are many bytes consist of three fields

Note this table is also included in options column. options may vary

Subfield name	**Size (bytes)**	**Description**
Option-kind	1	**Option-kind:** specifies the option type
Option-length:	1	**Option-length:** the length of the entire option in bytes, including the option-kind and option-length fields.
Option-data	Variable	**Option-data:** the option data itself in at least one oddbll case, this field is omitted(making option length equal to 2)

FINALTERM EXAMINATION

Fall 2008

CS610- Computer Network

Question No: 1 (Marks: 1) - Please choose one

In Direct point to point communication adding the Nth computer requires---------- new connections.

► None of the given

► N2

► **N-1**

► (N2 –N)/2

disadvantage In Direct point to point communication is that adding a new computer to the network requires N-1 new connections .(P# 23)

Question No: 2 (Marks: 1) - Please choose one

In Point-to-Point topology there are two topologies.

► Tree and Ring

► Star and Ring

► **Star and Tree**

► None of the given

POINT-TO-POINT:

In *Point-to-Point* topology there are two topologies.

1) STAR topology

2) TREE topology

Question No: 3 (Marks: 1) - Please choose one
In ----------, network occupies the smaller area like a room a floor or a building
► LAN
► WAN
► MAN
► None of the given
In LAN network occupies the smaller area like a room a floor or a building.(P# 4)
Question No: 4 (Marks: 1) - Please choose one
Hardware that calculates a CRC uses two simple components.
► AND unit and XOR unit
► Shift register and XOR unit
► Shift register and AND unit
► None of the given
CRC uses just two hardware components:
Shift register
Exclusive OR (XOR unit) .*(P# 20)*
Question No: 5 (Marks: 1) - Please choose one
CRC can detect more errors than a simple checksum.
► true
► false
Question No: 6 (Marks: 1) - Please choose one
The Gigabit Ethernet hardware operates at a rate of --------
► 10 Mbps
► 100 Mbps
► 1000 Mbps
► None of the given
Gigabit Ethernet that operates at 1 Gbps (1000 Mbps) over fiber optics and twisted pair Ethernet cables.
Question No: 7 (Marks: 1) - Please choose one
Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.
► 10 Base 2
► 10 Base 5
► 10 Base T
► None of the given
10BASE-T:
This is another standard of wiring scheme. It is commonly called 10Base-T, Twisted Pair or TP Ethernet. (P# 43)
Question No: 8 (Marks: 1) - Please choose one
An interface for thin Ethernet must have an _____ connector , and must generate signals according to the_____ specification.
► RJ-45, 10 Base T
► RJ-45, 10 Base 5
► BNC, 10 Base 2
► BNC, 10 Base T
Question No: 9 (Marks: 1) - Please choose one
A system with redundant bridges might have a problem with_____ in the system.
► Loop
► Filters

► Spanning Trees
► All given choices

Question No: 10 (Marks: 1) - Please choose one

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

► Greedy algorithm
► Distance vector algorithm
► Dijksta's algorithm
► Non of the given

Djikstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight (sum of the weight with all edges). It should be noted that the shortest path is not necessarily with fewest edges (or hops).

Question No: 11 (Marks: 1) - Please choose one

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

► Connectionless service paradigm
► Connection-oriented service paradigm
► Both Connectionless and Connection-oriented service paradigm
► None of the given

Question No: 12 (Marks: 1) - Please choose one

_____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer
► Network Interface Layer
► Internet Layer
► Transport Layer

LAYER 4: ensure reliable transfer layer 4 in TCP layering model is transport layer.

Question No: 13 (Marks: 1) - Please choose one

An Internet Address (IP address) is a unique _____ binary number assigned to a host and used for all communication with host

► 48-bit
► 32-bit
► 24-bit
► None of the given

An Internet address (IP address) is a unique 32-bit binary number assigned to a host and used for all communication with the host

Question No: 14 (Marks: 1) - Please choose one

The address _____ identifies the physical network to which the computer is attached, while the _____ identifies an individual computer on that network.

► prefix , suffix
► suffix , prefix
► suffix , suffix
► None of the given

PREFIX:

It identifies the physical network to which the computers are attached.

SUFFIX:

It identifies an individual computer on the network.

Question No: 15 (Marks: 1) - Please choose one

_____ places the boundary between the first and second octets

► Class A
► Class B

► Class C
► Class D

Class A places the boundary between the first and second octets. Class B places the boundary between the second and third octets, and class C places the boundary between the third and fourth octets.

Question No: 16 (Marks: 1) - Please choose one

_____ places the boundary between the third and fourth octets.

► Class A
► Class B
► Class C
► Class D

Class A places the boundary between the first and second octets. Class B places the boundary between the second and third octets, and class C places the boundary between the third and fourth octets.

Question No: 17 (Marks: 1) - Please choose one

_____ field of header indicates whether a datagram is a fragment or a complete datagram.

► FLAGS
► FLAGMENT OFFSET
► IDENTIFICATION
► None of the given

FRAGMENTATION:

IP uses fragmentation i.e. datagrams can be split into pieces to fit in network with small MTU. Each fragment is an independent datagram. It includes all header fields. Bit in header indicates that the datagram is a fragment

Question No: 18 (Marks: 1) - Please choose one

_____ provides connectionless service.

► TCP
► UDP
► IP
► None of the given

UDP has the following characteristics.

• It is an end-to-end protocol. It provides application-to-application communication.
• It provides connectionless service.
• It is a Message-Oriented protocol.
• It uses best-effort delivery service.
• It follows arbitrary interaction.
• It is operating system independent

Question No: 19 (Marks: 1) - Please choose one

UDP and TCP are both_____ layer protocols

► Physical
► Data link
► Network
► Transport

Question No: 20 (Marks: 1) - Please choose one

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by_____

► IP
► None of the given
► TCP

► UDP

SERVICE PROVIDED BY TCP:

Following are the services provided by TCP:

• Connection-oriented service
• Point-to-point
• Complete reliability
• Full-duplex communication
• Stream interface
• Reliable connection startup
• Graceful connection shutdown

Question No: 21 (Marks: 1) - Please choose one

_____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer
►Network Interface Layer
► Internet Layer
► Transport Layer

LAYER 4: ensure reliable transfer layer 4 in TCP layering model is transport layer

Question No: 22 (Marks: 1) - Please choose one

_____identifies which application program on receiving computer should receive the data

► Logical address
► Source port
► Destination Port
► None of the given

FieldDestinationPort identifies which application program on receiving computer should receive the data. While field source port identifies the application programs that sent the data. (reference from E-books)

Question No: 23 (Marks: 1) - Please choose one

_____ identifies the application program that sent the data.

► DestinationPort
► Source port
► Logical address
► None of the given

FieldDestinationPort identifies which application program on receiving computer should receive the data. While field source port identifies the application programs that sent the data. (reference from E-books)

Question No: 24 (Marks: 1) - Please choose one

The Border Gateway Protocol (BGP) uses_____ for all communication

► UDP
► TCP
► Both UDP and TCP
► None of the given

BORDER GATEWAY PROTOCOL:

It is most popular Exterior Gateway Protocol in Internet. It has following characteristics:

"It provides routing among autonomous systems (EGP).
"It provides policies to control routes advertised.
"It uses reliable transport (TCP).

Question No: 25 (Marks: 1) - Please choose one

Which of the following protocols allows the sender and receiver to enforce polices.

► RIP
► OSPF
► BGP
► RIP and OSPF

Question No: 26 (Marks: 1) - Please choose one

_____ uses distance vector approach to define routing

► BGP
► OSPF
► RIP
► None of the given

ROUTING INFORMATION PROTOCOL (RIP):

It has the following characteristics:

"It is used for routing within an autonomous system (IGP).

"Hop Count Metric: RIP measures distance in network hops, where each network between the source and destination counts as a single hop.

"It uses UDP for all message transmissions.

"RIP is used over LAN. Version 1 of RIP uses hardware broadcast and version 2 allows delivery via multicast.

"It can be used to advertise default route propagation. An organization can use RIP to install a default route in each router.

"It uses distance vector algorithm.

Question No: 27 (Marks: 1) - Please choose one

ICMP message transport is acted upon by getting ICMP encrypted in IP.

► True
► False

I*CMP message transport is acted upon by getting ICMP encapsulated in IP (P# 117)*

Question No: 28 (Marks: 1) - Please choose one

Protocol addresses are abstractions provided by _____.

► hardware
► software
► operating system
► internet

Protocol addresses are abstractions provided by software

Question No: 29 (Marks: 1) - Please choose one

These packets serve same purpose on _____ as frames on _____

► Intranet, LAN
► Internet, WAN
► Intranet, WAN
► Internet, LAN

VIRTUAL PACKETS:

These packets serve same purpose in Internet as frames on LAN

Question No: 30 (Marks: 1) - Please choose one

Address mask defines how many bits of address are in suffix?

► True
► False

Address mask defines how many bits of address are in prefix. Prefix defines how much of address used to identify network.

Question No: 31 (Marks: 1) - Please choose one

A single networking technology is best for all needs.
► True
► False
There is no single networking technology that is best for all needs
Question No: 32 (Marks: 1) - Please choose one
A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?

► True
► False
The chief problems with multiple networks are as follows:
• A computer attached to a given network can only communicate with other computers attached to the same network.
Question No: 33 (Marks: 1) - Please choose one
The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address_____.
► itself
► prefix
► suffix
► mask
Classful IP addresses are self-identifying because the class of the address can be computed from the address itself
Question No: 34 (Marks: 1) - Please choose one
Find the class of the address.
10100111 11011011 10001011 01101111
► A
► B
► E
► C
Reference table handouts P # 87
Question No: 35 (Marks: 1) - Please choose one
Find the class of the address:
11110011 10011011 11111011 00001111
► A
► C
► E
► B
Reference table handouts P # 87
Question No: 36 (Marks: 1) - Please choose one
In which method of Address Resolution Protocol the protocol address is determined by hardware address?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?
► T
► D
► C
► T, C
Reference table handouts P # 97
Question No: 37 (Marks: 1) - Please choose one

Which method of Address Resolution Protocol requires hardware broadcast?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► D

► T

► C

► T, D

Reference table handouts P # 97

Question No: 38 (Marks: 1) - Please choose one

Which method of Address Resolution Protocol resolution with minimum delay?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T, D

► c

► T

► T, C

Reference table handouts P # 97

Question No: 39 (Marks: 1) - Please choose one

In which method of Address Resolution Protocol the implimentation is more difficult?
Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T, C

► T

► C

► D

Reference table handouts P # 97

Question No: 40 (Marks: 1) - Please choose one

On of the design goals for unicast route propagation is _____.

► Consistency

► inconsistency

► stability

► dynamic addressing

On of the design goals for unicast route propagation is stability—continual changes in route are undesirable because they lead to higher jitter and datagram arriving out of order. Thus, once a unicast routing protocol finds a shortest path, it usually retains the route until a failure makes the path unusable. (Reference from E-book)

Question No: 41 (Marks: 1) - Please choose one

Propagation multicast routing information differs dramatically from unicast route propagation?

► True

► False

Propagating multicast routing information differs dramatically from unicast route propagation. The difference arises because internet multicast allows dynamic group membership and autonomous senders

Question No: 42 (Marks: 1) - Please choose one

The IP multicast abstraction allows an application running on an arbitrary computer to leave a multicast group at any time. While _____ application on a computer remain a member of a group.

► One or more

► only one

► no

► many

Question No: 43 (Marks: 1) - Please choose one
To save traffic, an EGP does not summerize routing information from the autonomous system before passing it to another autonomous system.
► True
► False
To save traffic, an EGP summerize routing information from the autonomous system before passing it to another autonomous system. More important an EGP implements policy constraints that allow a system manager to determine exactly what information is released outside the organization. (*Reference from E-book*)
Question No: 44 (Marks: 1) - Please choose one
In IPv6 the type of address used for collection of computers with same prefix. Are known as_____.
► Anycast
► Unicast
► Multicast
► Non of the given
Following are special types of addresses, IPv6 uses:
Unicast: *It is used for single destination computer.*
Multicast: *It is used for multiple destinations; possibly not at same site.*
Cluster: *This type of address is used for collection of computers with same prefix, datagram is delivered to one out of cluster.*
Question No: 45 (Marks: 1) - Please choose one
Special types of addresses in IPv6 used for multiple destinations; possibly not at same site. Are known as_____.
► Unicast
► Anycast
► Multicast
► Non of the given
Following are special types of addresses, IPv6 uses:
Unicast: It is used for single destination computer.
Multicast: It is used for multiple destinations; possibly not at same site.
Cluster: This type of address is used for collection of computers with same prefix, datagram is delivered to one out of cluster.
Question No: 46 (Marks: 1) - Please choose one
UDP offers application programs a Message-Oriented Interface, applications can depend on protocol to preserve data boundaries.
► True
► False
UDP offers application programs a Message-Oriented Interface. It does not divide messages into packets for transmission and does not combine messages for delivery.
ADVANTAGES:
• *Applications can depend on protocol to preserve data boundaries.*
Question No: 47 (Marks: 1) - Please choose one
Reliability is the responsibility of the _____ layer
► Network
► Datalink
► Transport
► Application

Reliability is the responsibility of the Transport layer

Question No: 48 (Marks: 1) - Please choose one

TCP uses _____ mechanism to control the flow of data.

► door

► window

► acknowledgment

► retransmission

TCP uses window mechanism to control the flow of data.

Question No: 49 (Marks: 1) - Please choose one

The time for acknowledgement to arrival of packet depends on.

► Distance to destination and Current traffic conditions

► Current traffic conditions

► Distance to destination

► non of these

***The time for acknowledgement to arrive depends on**:*

• *Distance to destination*

• *Current traffic conditions*

Question No: 50 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of ----------

► 100 million bits per second

► 10 million bits per second

► 1000 million bits per second

► None of the given

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are:

It uses fiber optics between stations and transmits data at 100Mbps.

Question No: 51 (Marks: 2)

How can we prove that we have 2,147,483,648 addresses in class A?

Question No: 52 (Marks: 2)

Why is internet multicast routing difficult?

Question No: 53 (Marks: 2)

Define what is the Extension Headers in IPv6.

Question No: 54 (Marks: 3)

How does IP software reassemble fragments that arrive out of order?

Question No: 55 (Marks: 3)

What is the first address in the block if one of the addresses is 167.199.170.82/27?

Question No: 56 (Marks: 3)

In internet routing how does a host join or leave a group?

Question No: 57 (Marks: 5)
What is the role of area in open shortest path first (OSPF)?
Question No: 58 (Marks: 5)
Write a note on Address Resolution.

Total Question 40
30 MCQs
70 % mcqs coming from attached file
some mcqs given below

1) When an application---------- data, it makes a copy of the data available to all other computers on the network.

► **Broadcasting**

► Multicasting

► Unicasting

► None of the given

2) Ethernet uses a ------ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

► 64

► **48**

► 32

► 8

3) The product of delay and throughput measures the _____ of data that can be present on the network.

► Area

► **Volume (supposition)**

► Length

► None of the given

4) Connectionless service, Message-Oriented protocol, best effort delivery service, arbitrary interaction and operating system independent are the characteristics of _____

► TCP

► **UDP**

► IP

► None of the given

5) _____ uses distance vector approach to define routing

► BGP

► OSPF

► **RIP**

► None of the given

6) Whenever it handles a packet, IP software needs to separate the destination address into a _____ and _____.

► postfix, Infix

► non of these

► Infix, prefix

► **prefix, suffix**

7) ARP is almost always used to bind a ___-bit IP address to a ___-bit Ethernet address.

► **32, 48**

► 24, 32

► 32, 64

► 32, 128

8) Which method of Address Resolution Protocol is useful with any hardware?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T

► C (Supposition)

► D

► C, D

9) In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T

► D

► C

► T, C

10)We use the term _____ to refer to a measure of the path that routing software use when choosing a route.

► routing path

► routing metric

► routing

► switching

11) A network uses a ------arranges for computers to be connected in a closed loop.

► Star Topology

► Ring Topology

► Bus Topology

► None of the given

12) An interface for thin Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

► RJ-45, 10 Base T

► RJ-45, 10 Base 5

► BNC, 10 Base 2

► BNC, 10 Base T

13) _____ protocols of TCP/IP layering model specify how to ensure reliable transfer.

► Physical Layer

► Network Interface Layer

► Internet Layer

► Transport Layer

14)_____ uses distance vector approach to define routing

► BGP

► OSPF

► RIP

► None of the given

4 question of 2 marks
i) question regarding Ip-multicasting (arbitrary computer to join multicast group)
ii) CRC detect what kind of error
iii) define IPV6 NEXT HEADER
iv) question regarding the metric of routing protocol
4 question of 3 marks
i) describe three feature of closed-form computation
ii) TCP timer for retransmission
iii) Difference between forwarding and routing

iv) Define ATM techonology

2 question of 10 marks

i) Define the CHARACTERISTICS OF UDP.

ii) question regarding the EGP (EXTERIOR GATEWAY PROTOCOL)